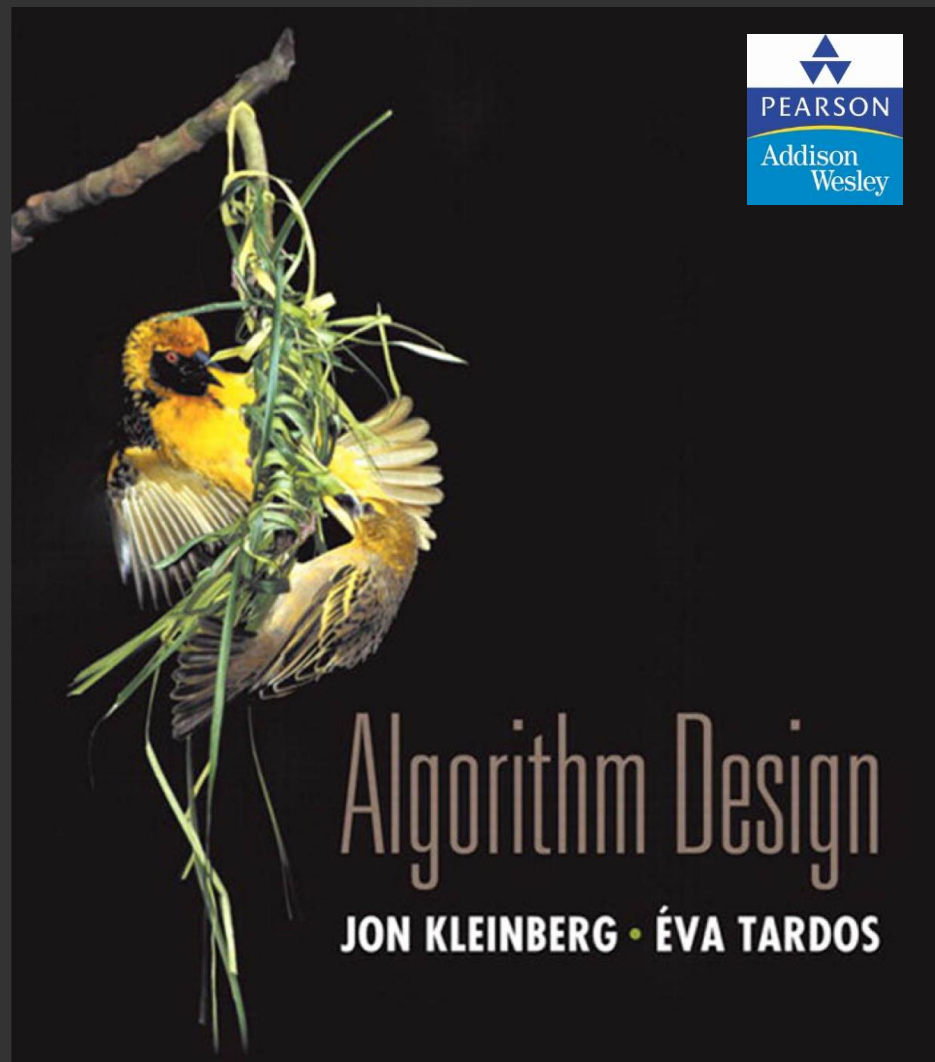




INTRACTABILITY III

- *greedy vertex cover algorithm demo*

Lecture slides by Kevin Wayne

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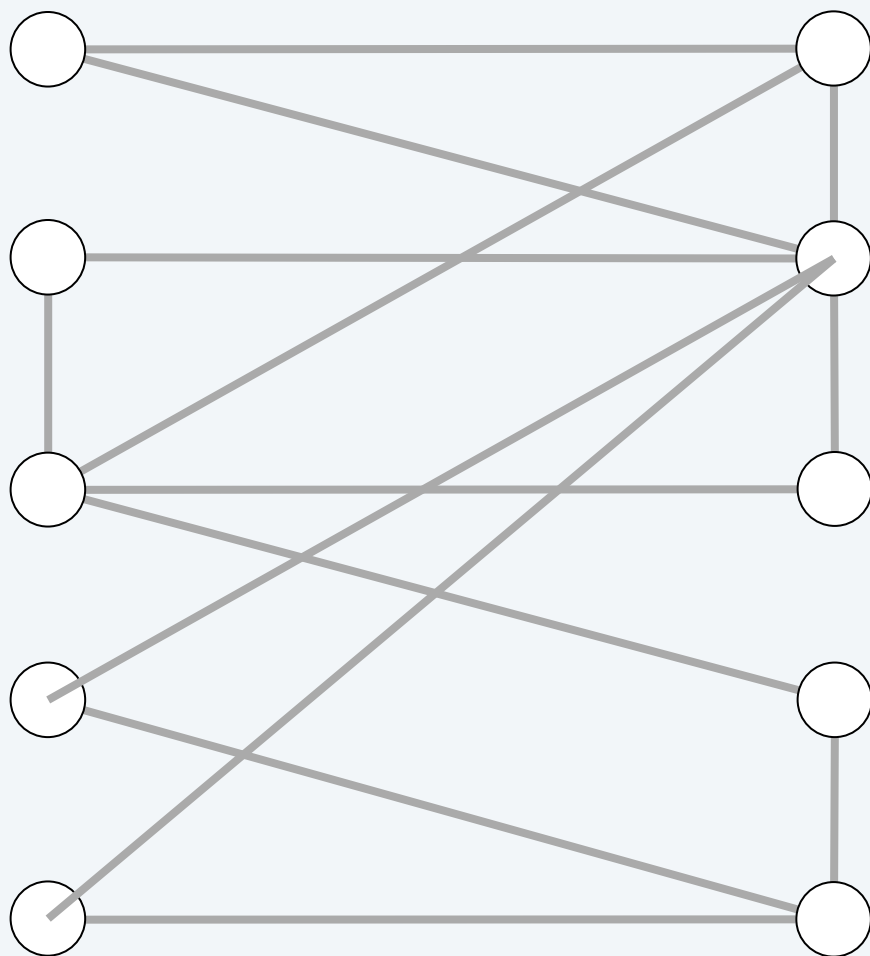
<http://www.cs.princeton.edu/~wayne/kleinberg-tardos>

Abdur Rashid Tushar
Lecturer, CSE, BUET

Greedy vertex cover algorithm

Repeatedly pick an arbitrary edge (u, v) .

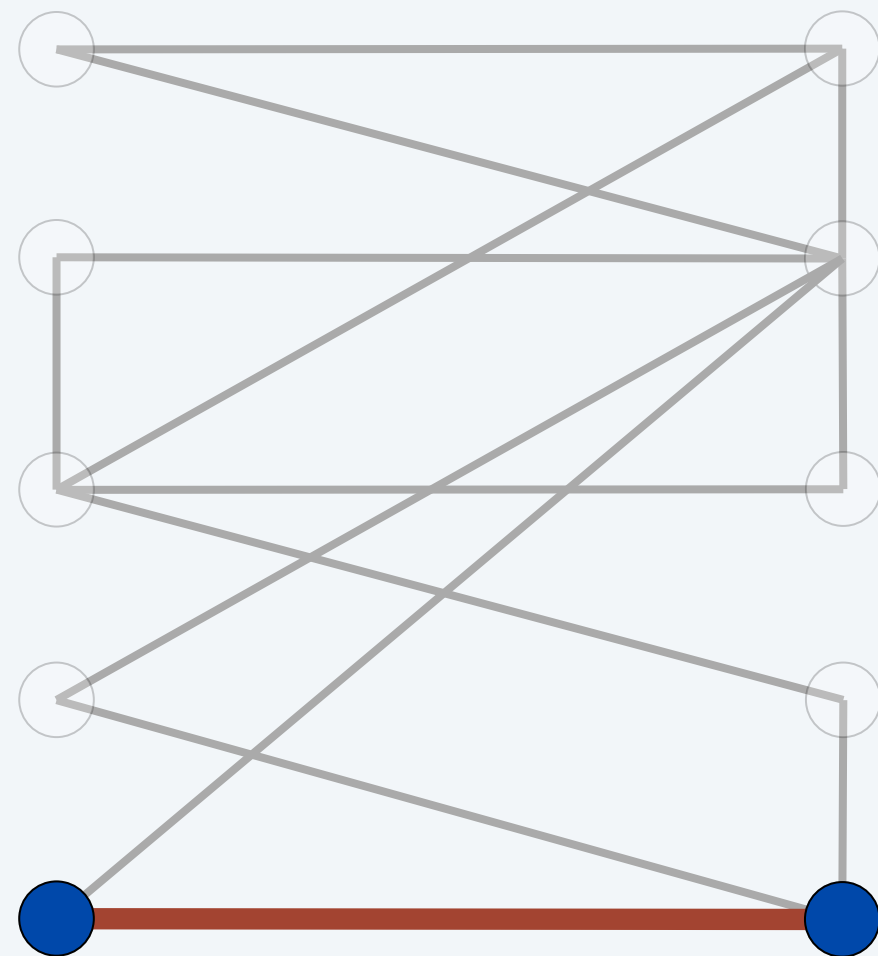
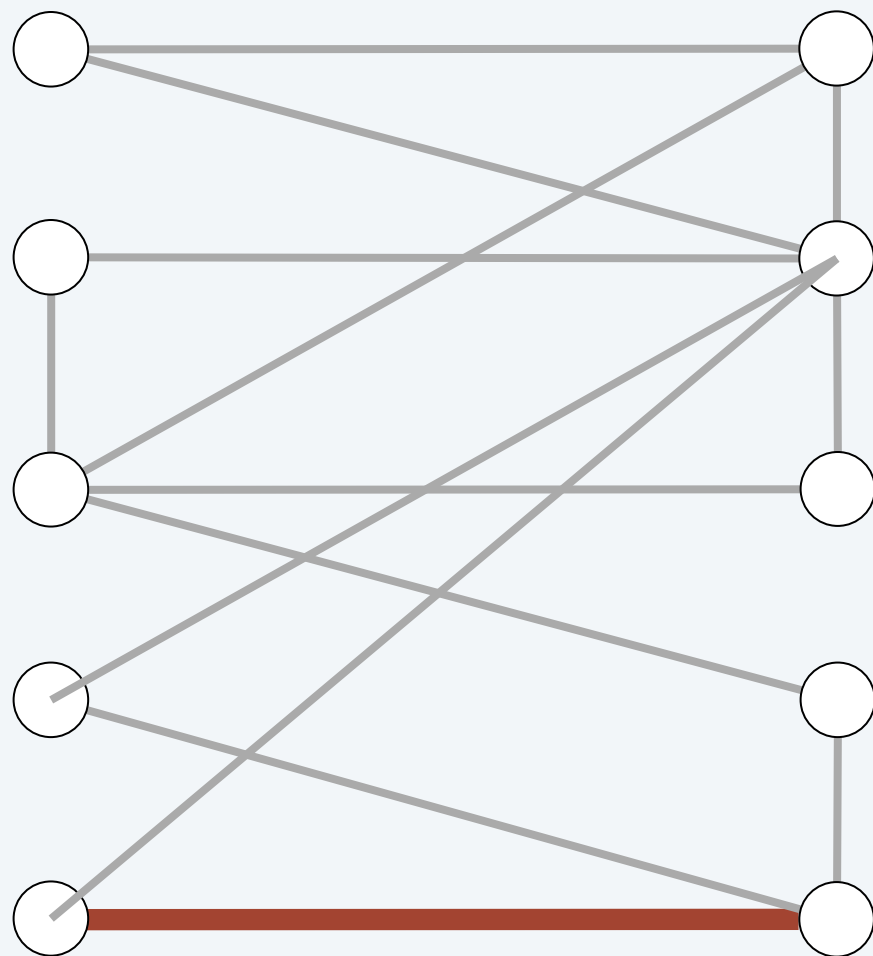
- Add (u, v) to matching M .
- Add both u and v to vertex cover S .
- Delete all edges incident to either u or v .



Greedy vertex cover algorithm

Repeatedly pick an arbitrary edge (u, v) .

- Add (u, v) to matching M .
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vertex cover S

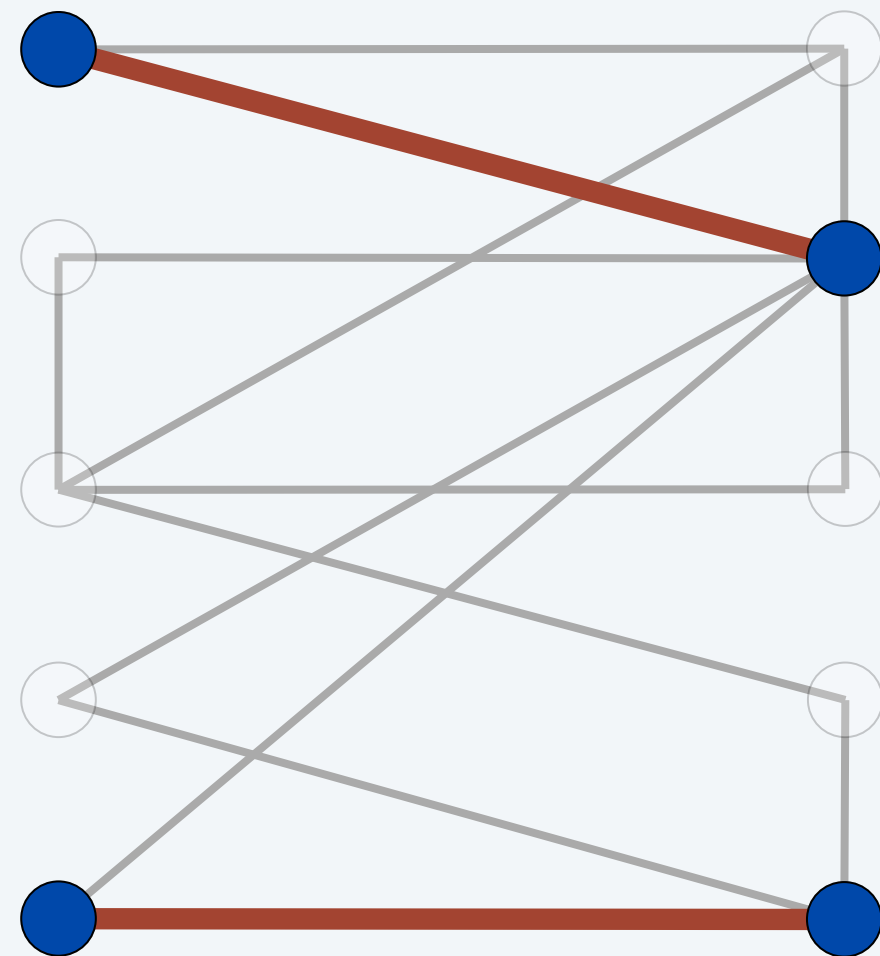
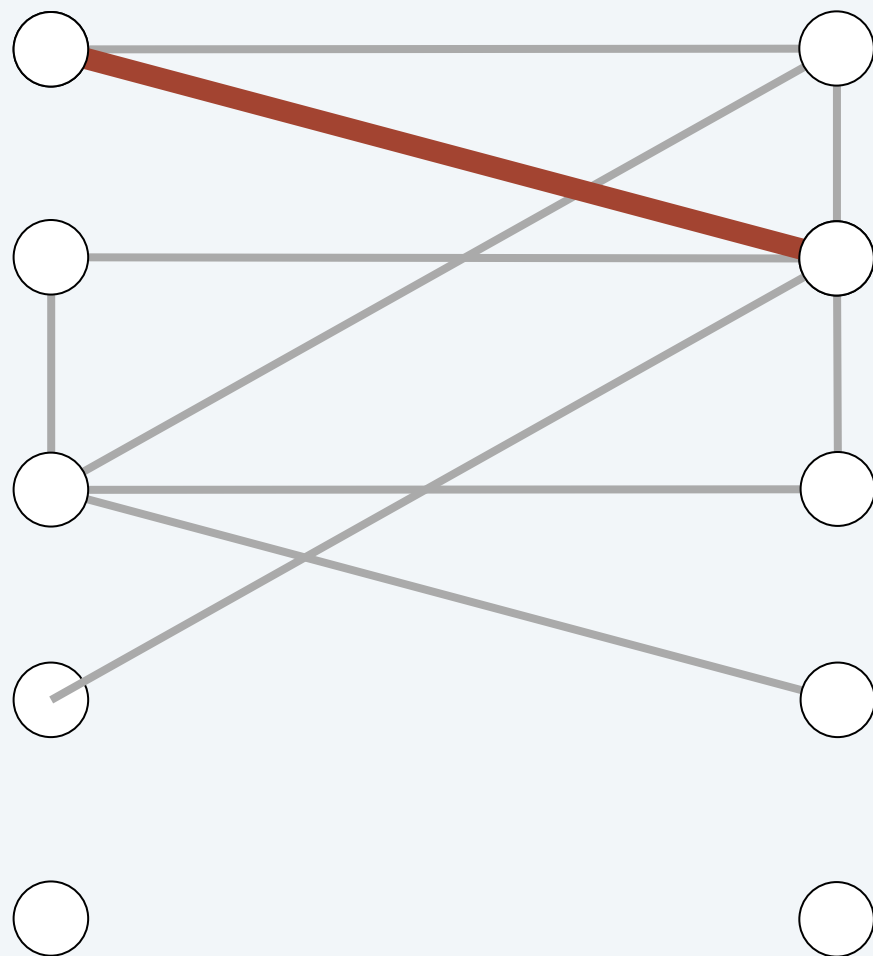


matching M

Greedy vertex cover algorithm

Repeatedly pick an arbitrary edge (u, v) .

- Add (u, v) to matching M .
- Add both u and v to vertex cover S .
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vertex cover S

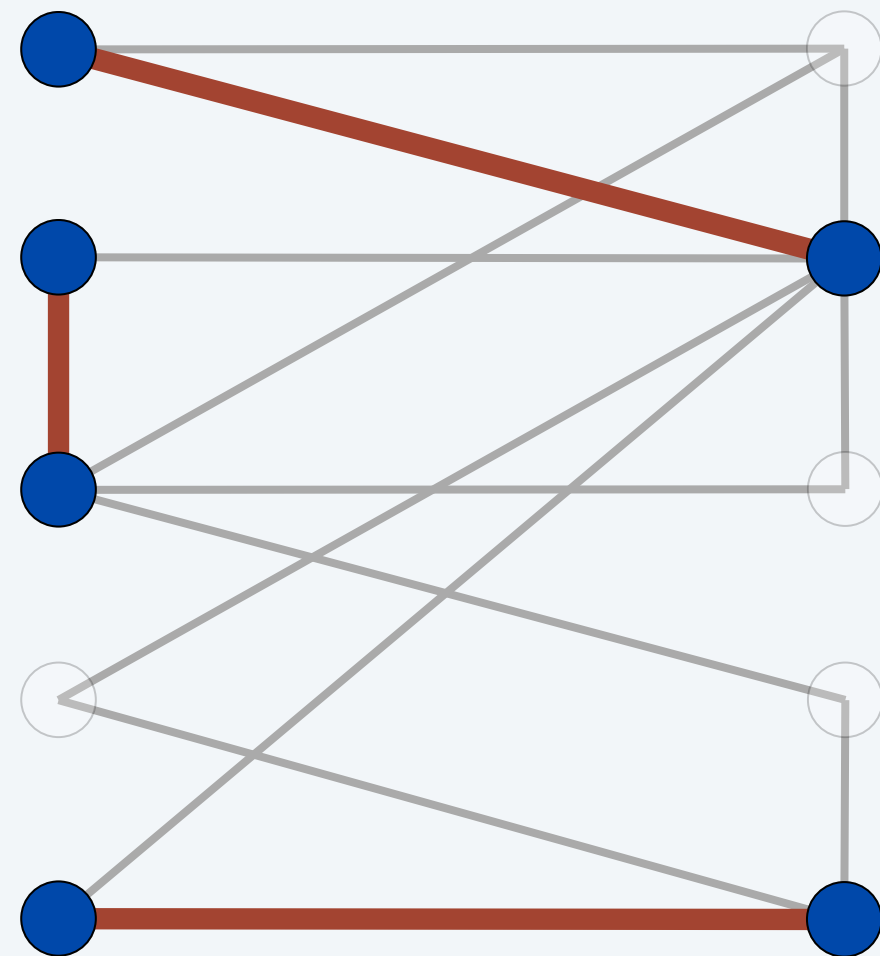
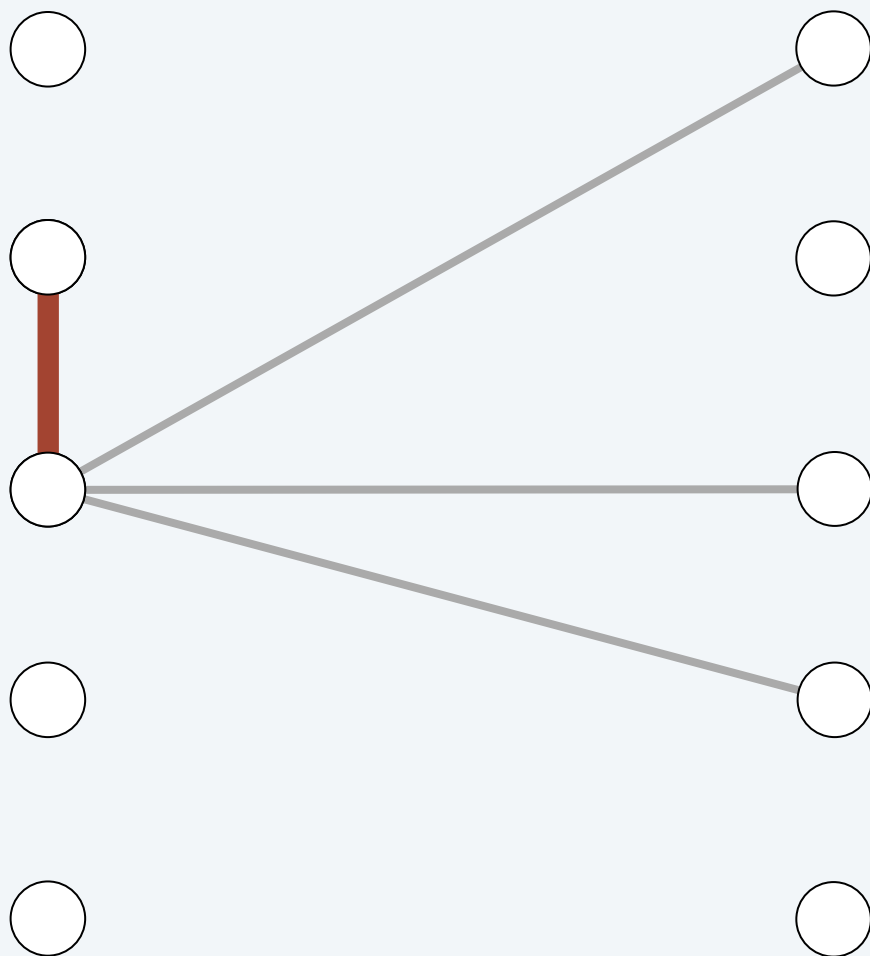


matching M

Greedy vertex cover algorithm

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vertex cover S

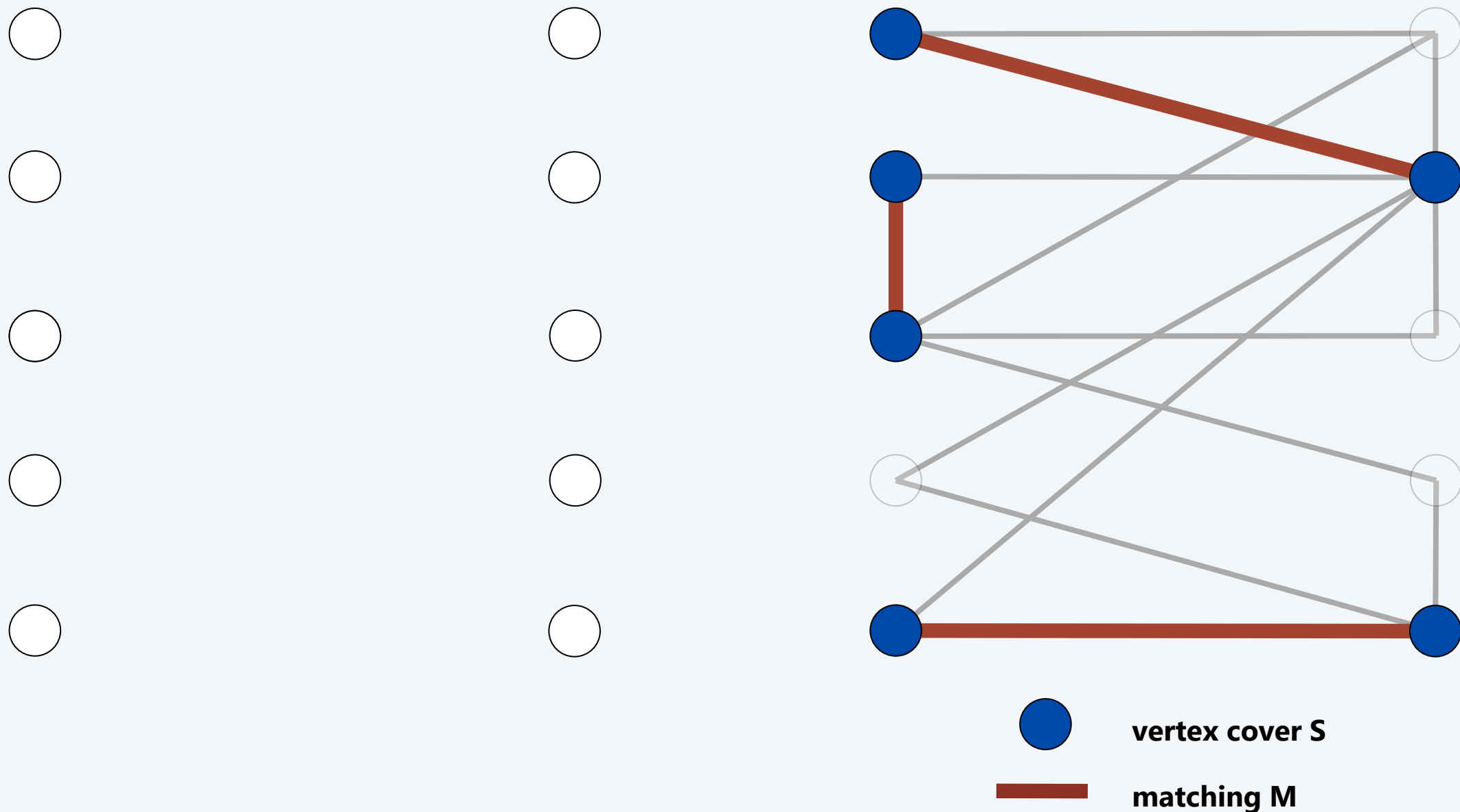


matching M

Greedy vertex cover algorithm

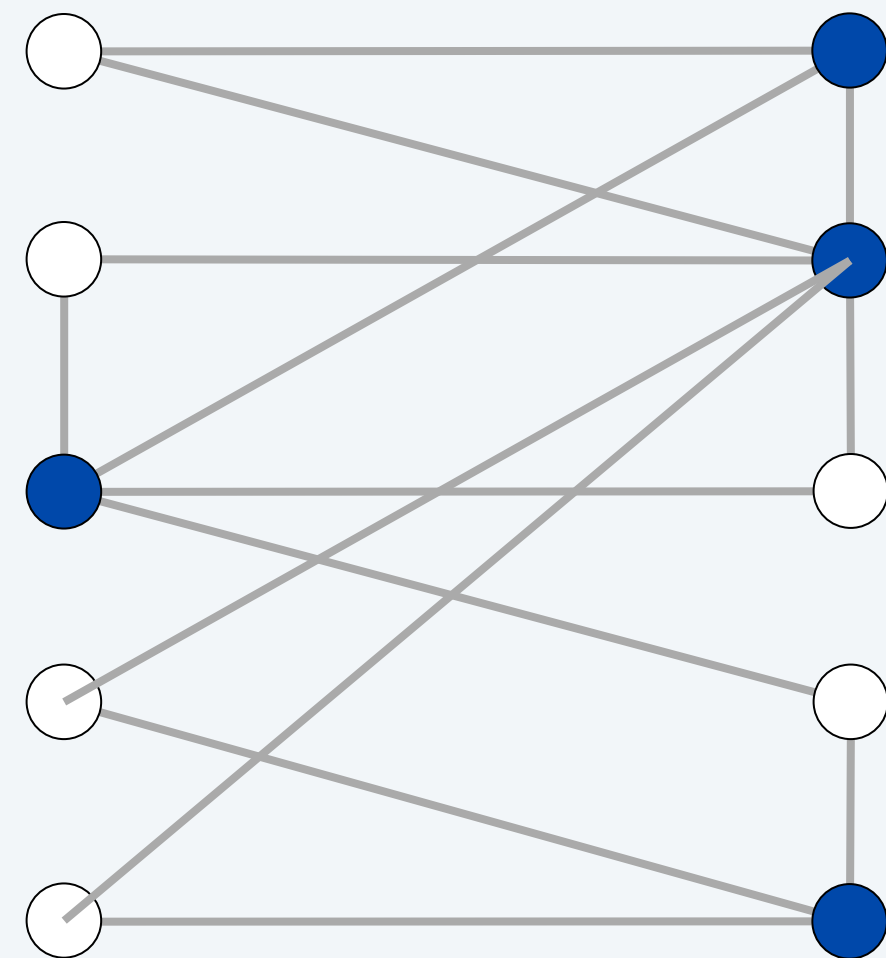
Repeatedly pick an arbitrary edge (u, v) .

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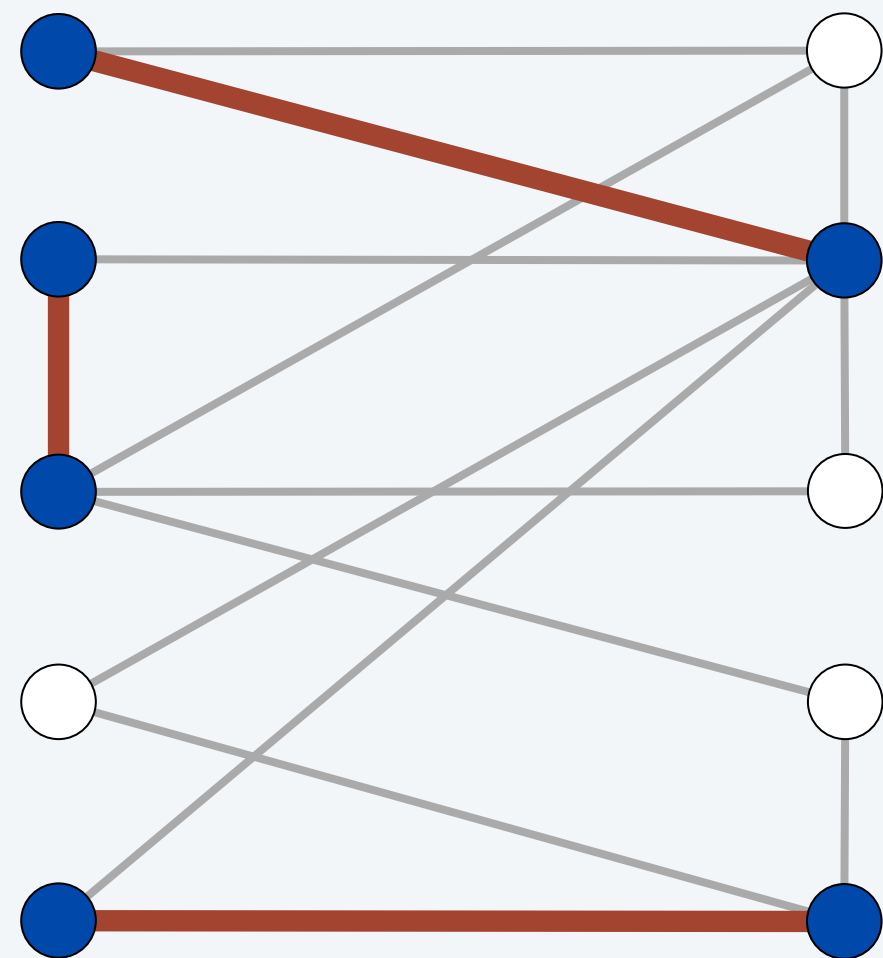


Greedy vertex cover algorithm

Greedy algorithm is not optimal.



● min-cardinality vertex cover S^*



● vertex cover S
— matching M